

A Strictly Positive Functional Subcase for a Kothe-Space Fremlin Tensor Representation Question

Automated Math Discovery Smoke Test

June 28, 2026

Source and Open Question

This packet concerns the function-space representation remark in Section 4 of:

Qingying Bu, Gerard Buskes, Alexey I. Popov, Adi Tcaciuc, and Vladimir G. Troitsky,
The 2-concavification of a Banach lattice equals the diagonal of the Fremlin tensor square, arXiv:1109.5380.

On page 13 the source paper states:

For $x, y \in E$, their elementary tensor product $x \otimes y$ can be viewed as a function on Ω^2 via $(x \otimes y)(s, t) = x(s)y(t)$ for $s, t \in \Omega$. This way, $E \otimes E$ is a subset of $L_0(\Omega^2)$. We do not know whether $E \otimes_{|\pi|} E$ can still be viewed as a sublattice of $L_0(\Omega^2)$, but this is definitely the case in many important special cases.

Let D be the diagonal of Ω^2 , that is, $D = \{(s, s) : s \in \Omega\}$. Of course, the map $s \rightarrow (s, s)$ is a bijection between Ω and D , so that we can view D as a copy of Ω . For

In words, if E is a Kothe space on a sigma-finite measure space (Ω, Σ, μ) , then $E \otimes E$ is represented in $L_0(\Omega^2)$ by

$$(x \otimes y)(s, t) = x(s)y(t).$$

The authors say that they do not know whether the completed Fremlin tensor product $E \widehat{\otimes}_{|\pi|} E$ can still be viewed as a sublattice of $L_0(\Omega^2)$, although this holds in many important special cases.

Candidate Partial Result

Theorem 1. *Let E be a Kothe Banach lattice on a sigma-finite measure space (Ω, Σ, μ) . Suppose there is a measurable function $h > 0$ a.e. and a constant $C < \infty$ such that*

$$\int_{\Omega} |x(\omega)| h(\omega) d\mu(\omega) \leq C \|x\|_E \quad (x \in E).$$

Then the canonical representation

$$x \otimes y \longmapsto [(s, t) \mapsto x(s)y(t)]$$

extends to an injective lattice homomorphism

$$J : E \widehat{\otimes}_{|\pi|} E \longrightarrow L_0(\Omega^2, \mu \otimes \mu).$$

Consequently, under this strictly positive Kothe-dual functional hypothesis, $E \widehat{\otimes}_{|\pi|} E$ can be viewed as a vector sublattice of $L_0(\Omega^2)$ extending the usual elementary-tensor representation.

Proof Intuition

The obstacle in the source question is that the Fremlin norm completion could, a priori, create abstract lattice elements with no measurable representative on the original product space. A strictly positive continuous Kothe-dual functional prevents this. It gives an equivalent weighted product measure $h(s)h(t) d\mu(s)d\mu(t)$ on Ω^2 . The elementary product map is then not merely a map into L_0 ; it is a bounded lattice bimorphism into the honest Banach lattice $L_1(h \otimes h)$. Fremlin's universal property extends this map to the completed tensor product. Strict positivity plus Fremlin's positive-cone domination then rules out a kernel.

Proof

Define

$$d\nu(s, t) = h(s)h(t) d\mu(s)d\mu(t)$$

on Ω^2 . Since $h > 0$ a.e., the measures ν and $\mu \otimes \mu$ have the same null sets.

Consider the bilinear map

$$m : E \times E \longrightarrow L_1(\nu), \quad m(x, y)(s, t) = x(s)y(t).$$

It is positive, and it is a lattice bimorphism because

$$|m(x, y)| = m(|x|, |y|) \quad (x, y \in E).$$

Moreover,

$$\|m(x, y)\|_{L_1(\nu)} = \int_{\Omega^2} |x(s)y(t)| h(s)h(t) d\mu(s)d\mu(t) = \left(\int_{\Omega} |x| h d\mu \right) \left(\int_{\Omega} |y| h d\mu \right) \leq C^2 \|x\|_E \|y\|_E.$$

Thus m is a continuous positive lattice bimorphism into the Banach lattice $L_1(\nu)$.

By the universal property of the Fremlin projective tensor product [2, 1E(iii), 1F], m induces a bounded positive operator

$$J : E \widehat{\otimes}_{|\pi|} E \longrightarrow L_1(\nu)$$

such that $J(x \otimes y) = m(x, y)$. Since m is a lattice bimorphism, the same universal property gives that J is a lattice homomorphism. Composing with the canonical inclusion $L_1(\nu) \subset L_0(\Omega^2)$ gives the desired extension into $L_0(\Omega^2)$.

It remains only to prove injectivity. Suppose $0 < U \in E \widehat{\otimes}_{|\pi|} E$. Fremlin's positive-cone domination property for the projective tensor product says that there are nonzero $x, y \in E_+$ with

$$0 < x \otimes y \leq U$$

in $E \widehat{\otimes}_{|\pi|} E$; see [2, 1(A)d]. Since x and y are nonzero positive functions and $h > 0$ a.e., the product $(s, t) \mapsto x(s)y(t)$ is nonzero in $L_1(\nu)$. Therefore

$$0 < J(x \otimes y) \leq J(U),$$

so $J(U) \neq 0$. Hence no nonzero positive element lies in the kernel of J .

Finally, if $J(W) = 0$ for an arbitrary $W \in E \widehat{\otimes}_{|\pi|} E$, then

$$J(|W|) = |J(W)| = 0.$$

The previous paragraph applied to the positive element $|W|$ gives $|W| = 0$, and hence $W = 0$. Thus J is injective.

Verification Notes

No computational verifier was needed. The proof is a direct universal property argument. The most important checks for human review are:

- the weighted L_1 estimate, which is exactly where the continuous strictly positive Kothe-dual functional is used;
- the fact that the pointwise product map is a lattice bimorphism;
- the citation to Fremlin’s positive-cone domination property, used to prove that the induced lattice homomorphism has trivial kernel.

Limitations and Novelty Check

This is a positive subcase, not a complete answer to the source question as stated. The packet assumes the existence of a strictly positive continuous Kothe-dual functional h . This hypothesis holds in the standard examples where such an associate weight is available, but the packet does not prove that every Kothe space in the minimal source-paper definition has this property.

The bounded literature check searched the run indexes for arXiv:1109.5380, “2-concavification”, “Fremlin tensor”, “Kothe”, and L_0 product phrases. It also checked the later related papers arXiv:1301.0749, arXiv:2404.02851, and arXiv:2601.19661. The 2013 Troitsky–Zabeti paper extends the diagonal/concavification theorem, while the 2024 Zabeti paper gives representation results in $S(X \times Y)$ -type spaces for order continuous Banach lattices. Neither source was found to give this exact same- Ω^2 , weighted- L_1 sufficient condition as a resolution of the source remark. Web searches on June 28, 2026 for exact phrases from the remark and for Fremlin tensor products of Banach function/Kothe spaces did not reveal an explicit full answer to the unrestricted question.

References

References

- [1] Q. Bu, G. Buskes, A. I. Popov, A. Tcaciuc, and V. G. Troitsky, *The 2-concavification of a Banach lattice equals the diagonal of the Fremlin tensor square*, arXiv:1109.5380.
- [2] D. H. Fremlin, *Tensor products of Banach lattices*, Math. Ann. **211** (1974), 87–106.
- [3] V. G. Troitsky and O. Zabeti, *Fremlin tensor products of concavifications of Banach lattices*, arXiv:1301.0749.
- [4] O. Zabeti, *A representation theorem for order continuous Banach lattices with applications*, arXiv:2404.02851.